

# Evaluation of Start-Up Performance of Induction Motor Speed Sensorless Vector Control Using Adaptive Flux Observer Considering Inverter Voltage Error

Erina Izumi<sup>\*a)</sup> Student Member,  
 Keiichiro Kondo<sup>\*</sup> Senior Member,  
 Hiromitsu Suzuki<sup>\*\*</sup> Member

Masaki Nagataki<sup>\*</sup> Member  
 Shunsuke Tobayashi<sup>\*\*</sup> Member

(Manuscript received Jan. 00, 20XX, revised May 00, 20XX)

Speed sensorless vector control for induction motors using an adaptive flux observer makes it difficult to output high and stable torque at a standstill condition when the output voltage error mainly due to dead-time of the inverter. This voltage error leads to a speed estimation error. Consequently, axis misalignment occurs, and a torque value that tracks the reference command value cannot be output. A method has been proposed to compensate for this voltage error by using a current command value; however, the compensation accuracy is limited because of the forward voltage drop and nonlinear characteristics of power switching devices. This study applies the conventional dead-time compensation to the model reference adaptive system, and the effect of varying the amount of compensation, instead of investigating the true dead-time length, on the start-up is evaluated using a 2.2 kW experimental system and through dynamic analysis.

**Keywords:** speed-sensorless control, vector control, induction motor (IM), MRAS, dead-time, adaptive flux observer

## 1. Introduction

Speed sensorless vector controls of induction motors (IMs) are widely applied to general-purpose inverters, industrial drive systems, and railway vehicle traction for reducing motor size and maintenance costs, and increasing reliability by eliminating the influence of sensor breakdown<sup>(2)-(4)</sup>. However, the torque or speed control performance is highly sensitive to input voltage errors in the ultralow speed range, including the start-up from a standstill condition. Sensorless vector control by estimating the direction of the secondary flux through the secondary induced voltage<sup>(2)</sup> or primary q-axis current<sup>(3)</sup> is often used because of the low computational load on microprocessors. These methods assume that a secondary flux is established within a machine. However, at very low speeds, the voltage is too low to establish a correct secondary flux vector. Subsequently, the error in the induced voltage becomes relatively large; therefore, consider the dynamics of the magnetic flux is important in speed-estimation tasks. Therefore, another method to determine the primary frequency must be combined with an ultralow-speed range operation. For certain applications, a seamless control method is preferable. In such cases, speed-sensorless vector control with the model reference adaptive system

(MRAS) theory<sup>(4)</sup> is among the best options because it uses an adaptive secondary flux observer that considers the time derivatives of the secondary flux and primary current. This method can also be combined with systematic parameter-error compensation methods<sup>(5)(6)</sup>.

However, in the ultralow speed range, this method is strongly affected by the inverter's dead-time and primary voltage errors owing to switching devices, forward voltage drops, nonlinear characteristics, and other factors. The primary voltage errors reduce the voltage applied to the actual machine and introduce errors in the actual machine and adaptive flux observer inputs, causing errors in the estimation of the primary current and secondary flux in the adaptive flux observer. These errors degrade the rotor speed estimation performance and cause a misalignment in the  $dq$  coordinates owing to the secondary magnetic flux, resulting in a loss of the desired torque output.

There have been several discussions on the voltage errors caused by inverters. The conventional method of compensating for dead-time requires the detection of stator current polarity. During compensation, delays in polarity detection and compensation voltage mismatching cause large distortions in the current waveforms and torque ripple<sup>(7)</sup>. At ultralow speeds, the polarity transition is particularly slow, which makes detecting the polarity of the load current difficult. Compensating for nonlinear characteristics is also difficult.

Reference (7) proposed a method for compensating for nonlinear errors without current polarity discrimination. This method is characterized by use of a disturbance observer to detect voltage errors.

This paper is based on Reference (1), which published in the International Power Electronics Conference (2022) ©2022 IEEJ.

a) Correspondence to: Erina Izumi. E-mail:

s60204ei@fuji.waseda.jp

\* Waseda University

3-4-1, Okubo, Shinjuku-ku, Tokyo 169-8555, Japan

\*\* Toshiba Mitsubishi-Electric Industrial Systems Corporation  
 1, Toshiba-cho, Fuchu-shi, Tokyo 183-8511, Japan

Reference (8) proposed a method which uses the trapezoidal voltage as the compensation voltage. In this method, the modulation shape of the trapezoidal voltage mitigates the undesirable harmonic distortions in the overall system.

A compensation method focusing on the effect of dead-time in the zero-crossing zone of the stator current was also proposed in reference (9).

In reference (10), a look-up table is created and the voltage is compensated using the current estimation error.

In all methods from reference (7) to (10), the nonlinear error voltage is compensated by feedback (FB) control. However, the structure is simpler and the computational burden is smaller without FB control. In this study, conventional dead-time compensation by feed-forward (FF) control using current commands without any special control was used, and the effect of misalignment due to the nonlinear error of the inverter voltage errors was evaluated. As nonlinear voltage error amount cannot be determined, the amount of dead-time compensation is varied to reproduce the relative misalignment in compensation owing to nonlinear error. In such a case, whether overcompensation or undercompensation can provide more stable operation was investigated through experiments and dynamic characteristic analysis. The results of the overcompensation method produce unstable poles in the analysis and oscillations in the adaptive flux observer's estimated current and magnetic flux, as well as in the speed estimation that occurred in the experiments, making the entire system oscillatory and that the undercompensation is more stable to start-up.

## 2. Adaptive Flux Observer

Fig. 1 shows the configuration of the control system. Fig. 2 shows the configuration of the adaptive flux observer. The state equation is expressed by the following equations with respect to the

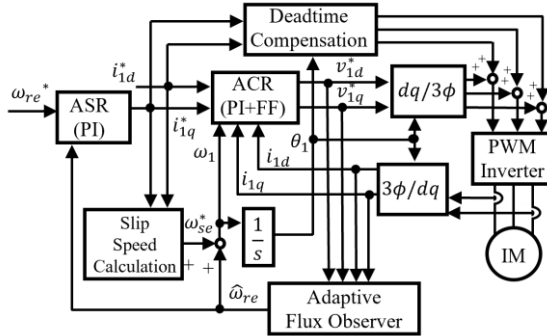


Fig. 1. Configuration of sensorless control

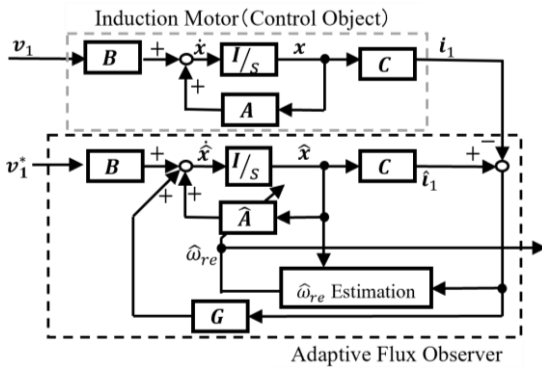


Fig. 2. Configuration of adaptive flux observer

secondary flux-oriented rotating  $dq$  coordinates, with the primary current  $i_1$  and secondary flux  $\phi_2$  of the induction motor as the state variables<sup>(11)</sup>.

$$p \begin{bmatrix} i_1 \\ \phi_2 \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} i_1 \\ \phi_2 \end{bmatrix} + \begin{bmatrix} B_1 \\ 0 \end{bmatrix} v_1 \quad (1)$$

$$y = i_1 = Cx \quad (2)$$

where,

$$i_1 = \begin{bmatrix} i_{1d} & i_{1q} \end{bmatrix}^T : \text{primary current}, \quad \phi_2 = \begin{bmatrix} \phi_{2d} & \phi_{2q} \end{bmatrix}^T : \text{secondary}$$

$$\text{flux}, v_1 = \begin{bmatrix} v_{1d} & v_{1q} \end{bmatrix}^T : \text{primary voltage}$$

$$A_{11} = -\left\{ \frac{R_1}{\sigma L_1} + \frac{R_2(1-\sigma)}{\sigma L_2} \right\} I - \omega_1 J = a_{r11} I + a_{i11} J$$

$$A_{12} = \frac{M}{\sigma L_1 L_2} \left( \frac{R_2}{L_2} I - \omega_{re} J \right) I = a_{r12} I + a_{i12} J$$

$$A_{21} = -\frac{MR_2}{L_2} I = a_{r21} I$$

$$A_{22} = -\frac{R_2}{L_2} I - \omega_{se} J = a_{r22} I + a_{i22} J$$

$$B_1 = \frac{1}{\sigma L_1} I = b_1 I$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$\sigma = 1 - \frac{M^2}{L_1 L_2} : \text{leakage coefficient}$$

$\omega_r$  : secondary angular speed (mechanical angle)

$\omega_{re}$  : secondary angular speed (electric angle)

$\omega_{se}$  : slip angular speed (electric angle)

$\omega_1$  : primary angular speed

$p$  : differential operator

The relevant parameters are listed in Table 2. According to equation (1), the adaptive flux observer used for state variable estimation is configured as follows:

$$p \begin{bmatrix} \hat{i}_1 \\ \hat{\phi}_2 \end{bmatrix} = \begin{bmatrix} \hat{A}_{11} & \hat{A}_{12} \\ \hat{A}_{21} & \hat{A}_{22} \end{bmatrix} \begin{bmatrix} \hat{i}_1 \\ \hat{\phi}_2 \end{bmatrix} + \begin{bmatrix} \hat{B}_1 \\ 0 \end{bmatrix} v_1^* + G(i_1 - \hat{i}_1) \quad (3)$$

Symbols  $\hat{\cdot}$  and  $\cdot^*$  indicate the estimated and command values, respectively.  $v_1^*$  denotes the primary voltage of the adaptive flux observer and uses the voltage command output value of the automatic current regulator (ACR). To ensure stable operation over a wide speed control range, the adaptive flux observer poles are set as  $k$  ( $> 0$ ) times higher than the poles of the original IM model. The adaptive flux observer gain matrix is given by the following equation<sup>(11)</sup>.

$$G = \begin{bmatrix} g_1 & g_2 & g_3 & g_4 \\ -g_2 & g_1 & -g_4 & g_3 \end{bmatrix}^T$$

where,

$$g_1 = (1-k)(\hat{a}_{r11} + \hat{a}_{r22})$$

$$g_2 = (1-k)(\hat{a}_{i11} + \hat{a}_{i22})$$

$$g_3 = \frac{\hat{a}_{r12}\hat{g}_1 + (\hat{a}_{r12}\hat{a}_{r22} + \hat{a}_{i12}\hat{a}_{i22})g_1 + \hat{a}_{i12}\hat{g}_2 - (\hat{a}_{r12}\hat{a}_{i22} - \hat{a}_{i12}\hat{a}_{r22})g_2}{\hat{a}_{r12}^2 + \hat{a}_{i12}^2}$$

$$g_4 = \frac{-\hat{a}_{i12}\hat{g}_1 + (\hat{a}_{r12}\hat{a}_{i22} - \hat{a}_{i12}\hat{a}_{r22})g_1 + \hat{a}_{r12}\hat{g}_2 + (\hat{a}_{r12}\hat{a}_{r22} + \hat{a}_{i12}\hat{a}_{i22})g_2}{\hat{a}_{r12}^2 + \hat{a}_{i12}^2}$$

$$\tilde{g}_1 = (1-k^2)(-\hat{a}_{r11}\hat{a}_{r22} + \hat{a}_{i11}\hat{a}_{i22} + \hat{a}_{r12}\hat{a}_{r21})$$

$$\tilde{g}_2 = (1-k^2)(-\hat{a}_{r11}\hat{a}_{i22} + \hat{a}_{i12}\hat{a}_{i21} - \hat{a}_{i11}\hat{a}_{r22})$$

Speed estimation is calculated according to the following adaptive scheme <sup>(12)</sup>.

$$\hat{\omega}_{re} = K_{\hat{\omega}P} (e_{i1d}\hat{\phi}_{2q} - e_{i1d}\hat{\phi}_{2d}) + K_{\hat{\omega}I} \int (e_{i1d}\hat{\phi}_{2q} - e_{i1d}\hat{\phi}_{2d}) dt \dots\dots\dots (4)$$

where,  $e_{i1d} = i_{1d} - \hat{i}_{1d}$ ,  $e_{i1q} = i_{1q} - \hat{i}_{1q}$ ,  $K_{\hat{\omega}P}$ ,  $K_{\hat{\omega}I}$  : positive constant.

In vector control, the slip angular speed command value and primary angular speed are given by the following equations:

$$\omega_{se}^* = \frac{R_2 i_{1q}^*}{L_2 i_{1d}^*} \dots\dots\dots (5)$$

$$\omega_1 = \hat{\omega}_{re} + \omega_{se}^* \dots\dots\dots (6)$$

### 3. Dead-time Compensation Method

Fig. 3 shows the FF dead-time compensation method <sup>(13)</sup>. The polarity of actual voltage error due to dead-time is determined by the polarity of the real current. The value of the voltage error depends on the length of the dead time, the DC link voltage and the carrier frequency. Time average of the half cycle of the error voltage due to dead-time in the three phases has a magnitude of  $V_{DC} f_c T_{DT}$ , and it is considered to be square waves. In compensation, the current command values of the  $dq$  rotational axis are converted to the  $uvw$  3-phase command value and the polarities of those are detected and used for the compensation. In some cases, the feedback actual current is used for the compensation, but in this case, the current command value is used. The compensation amount of  $\Delta v$  is added to the 3-phase voltage command value, which is the output of the automatic current controller and PWM is performed.  $\Delta v$  is a magnitude of  $V_{DC} f_c T_{DT}$  in general. For changing the amount of dead-time compensation, this value is multiplied by a factor  $\alpha$  and adjusted as in equation (7).

$$\Delta v = V_{DC} f_c T_{DT} \alpha \dots\dots\dots (7)$$

If  $\alpha > 1.0$  : Over compensation,  $\alpha < 1.0$  : Under compensation

### 4. Start-up Experiment with the Dead-time Compensation

#### 4.1 The experimental set up

The start-up characteristics from a standstill condition with dead-time compensation were examined experimentally. Fig.4 illustrates the configuration of the experimental equipment. Secondary angular speed  $\omega_r$  is measured by detecting and differentiating the angle by an encoder installed on the same axis of the IM and PMSM. The value is multiplied by the number of pole pairs  $P_n$  to obtain the electric secondary angular speed  $\omega_{re}$ . The actual slip  $\omega_{se}$  is the

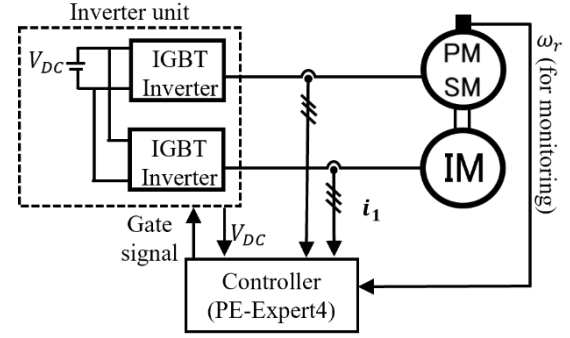


Fig. 4. System configuration used in the experiments

Table 1. Ratings of tested IM

|        |          |         |
|--------|----------|---------|
| 2.2 kW | 200 V    | 8.8 A   |
| 50 Hz  | 2860 rpm | 2 poles |

Table 2. Configuration of IM and control circuit

| Symbol              | Meaning                                                 | Value                     |
|---------------------|---------------------------------------------------------|---------------------------|
| $P_n$               | pole pair                                               | 1                         |
| $R_1$               | primary resistance                                      | 0.9000 $\Omega$           |
| $R_2$               | secondary resistance                                    | 0.5097 $\Omega$           |
| $L_1$               | primary self-inductance                                 | 0.1291 H                  |
| $L_2$               | secondary self-inductance                               | 0.1291 H                  |
| $M$                 | mutual inductance                                       | 0.1240 H                  |
| $J$                 | composite moment of inertia                             | 0.006282 kgm <sup>2</sup> |
| $\tau_e$            | rotational resistance                                   | 0.3320 Nm                 |
| $V_{DC}$            | DC link voltage                                         | 300 V                     |
| $f_c$               | carrier frequency                                       | 10 kHz                    |
| $T_{DT}$            | deadtime                                                | 3.0 $\mu$ s               |
| $T_c$               | control cycle                                           | 100 $\mu$ s               |
| $T_{ACR}$           | time constant of current controller                     | 10 ms                     |
| $K_{P(ACR)}$        | proportional gain of ACR ( $= \frac{\sigma L_1}{T_d}$ ) | 0.9999                    |
| $K_{I(ACR)}$        | integral gain of ACR ( $= \frac{R_1}{T_d}$ )            | 90                        |
| $K_{P(ASR)}$        | proportional gain of ASR                                | 0.2193                    |
| $K_{I(ASR)}$        | integral gain of ASR                                    | 0.8773                    |
| $K_{\hat{\omega}P}$ | proportional gain of $\omega_{re}$ estimation           | 1.0                       |
| $K_{\hat{\omega}I}$ | integral gain of $\omega_{re}$ estimation               | 2000                      |
| $k$                 | ratio of adaptive flux observer's poles to IM's poles   | 1.0                       |

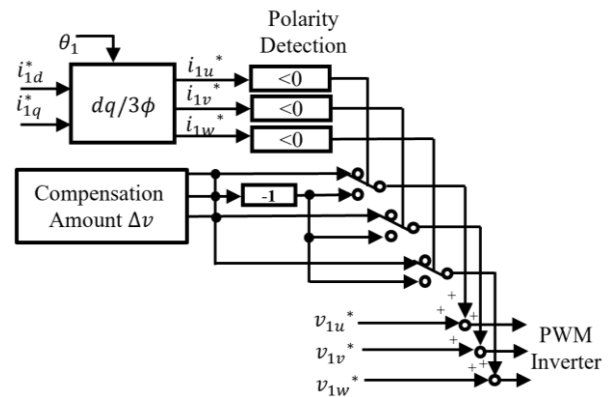


Fig. 3. Configuration of dead-time compensation

value obtained by subtracting the secondary angular speed  $\omega_{re}$  acquired by the sensor from the primary angular speed  $\omega_1$ . Table 1 lists the rating values for induction motors used in the experiments. The motor and control constants are listed in Table 2. A 2.2 [kW] IM driven by a speed control with an estimated speed under 100% of the rated load was used in the experiment. The load was emulated by a permanent magnet synchronous motor (PMSM), which was controlled by vector control with a speed sensor.

**4.2 Experimental results of start-up when dead-time compensation is misaligned** First, the d-axis current is maintained constant for a sufficient time to establish secondary flux at the standstill condition. The speed command value  $\omega_{re}^*$  is then input in ramp at a rate of 10[rad/s<sup>2</sup>]. As the induction motor starts from a standstill condition, the speed estimation value is assumed to be zero until the speed command input, state variables, primary current, and secondary flux are estimated by the adaptive flux observer using these variables. The motor parameters are assumed to be correct.

Fig. 5 shows the wave forms of experimental results without dead-time compensation ( $\alpha = 0.0$ ). The errors in the estimated primary current  $\hat{i}_1$  and secondary flux  $\hat{\phi}_2$  by the adaptive flux observer are misaligned significantly at low speeds and the estimated speed is also misaligned. This also causes slip mismatch and accelerating sufficiently because sufficient torque cannot be produced.

Fig. 6 shows the result of putting the dead-time compensation  $\Delta v$  shown in equation (7) at  $\alpha = 1.0$ .

The estimated primary current  $\hat{i}_1$  and secondary flux  $\hat{\phi}_2$  by the adaptive flux observer are close to the real primary current  $i_1$  and secondary flux command  $\phi_{2d}^*$  from low speeds, and the estimated

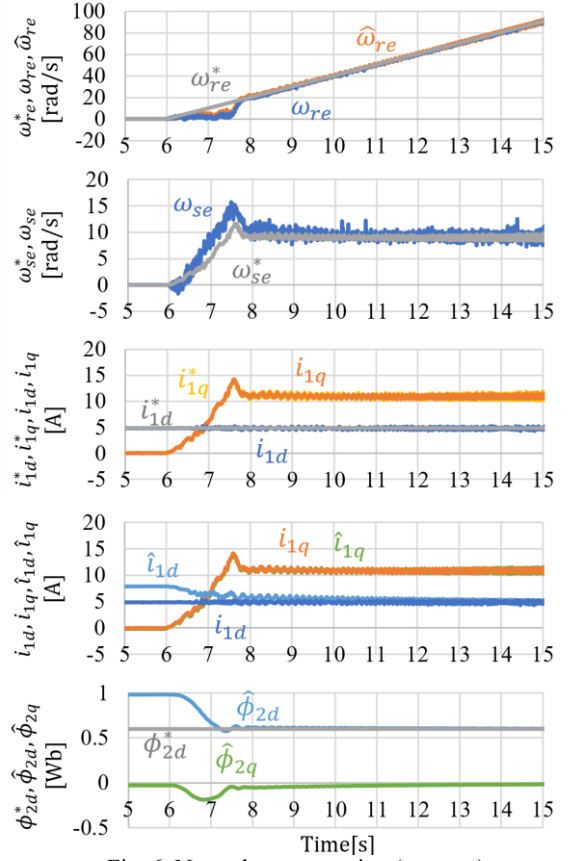


Fig. 6. Normal compensation ( $\alpha = 1.0$ )

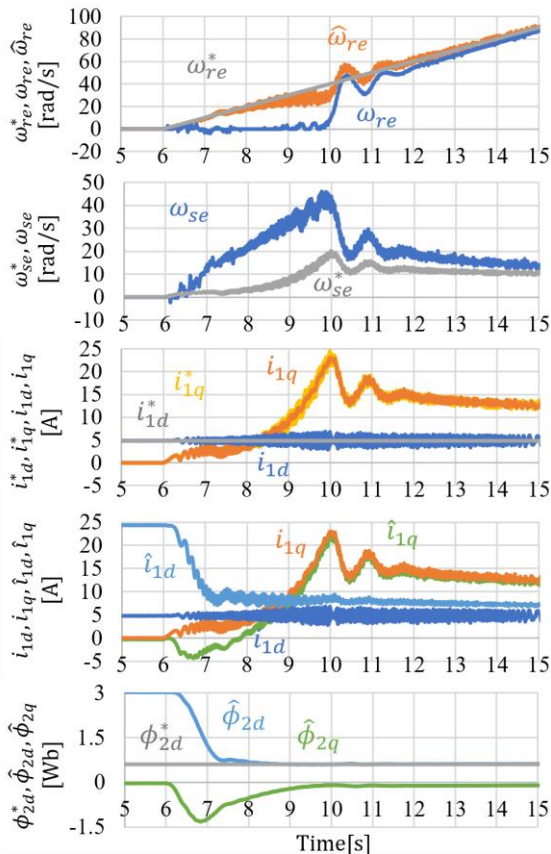


Fig. 5. Without dead-time compensation ( $\alpha = 0.0$ )

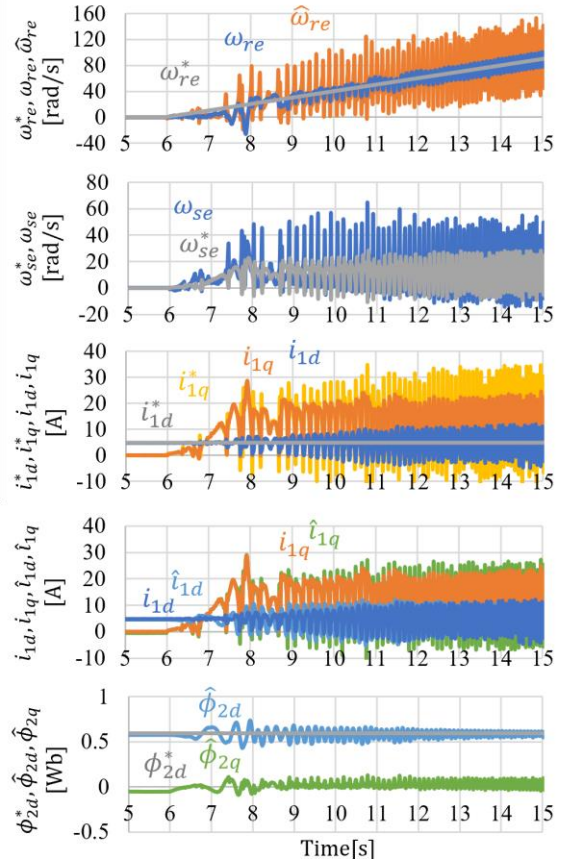
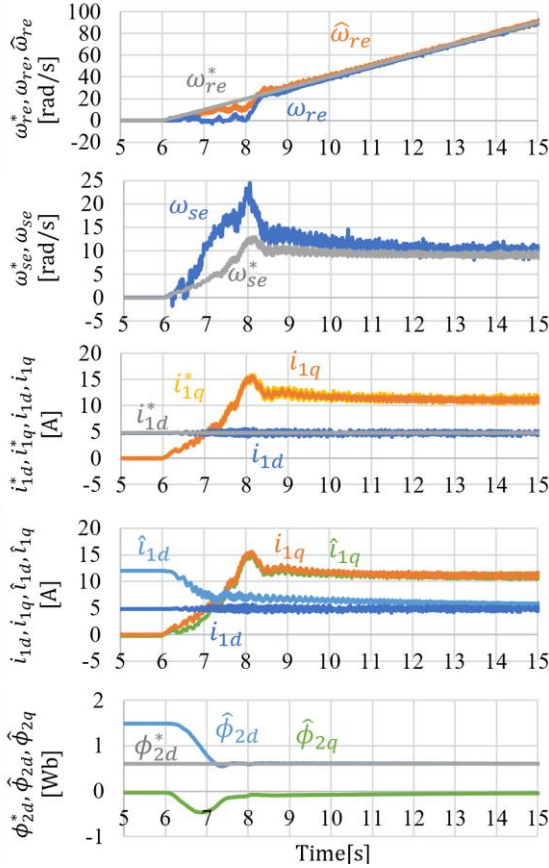


Fig. 7. Over compensation ( $\alpha = 1.25$ )

Fig. 8. Under compensation ( $\alpha = 0.75$ )

speed also matches the real speed from very low speeds.

It takes approximately 1.0[s] to start; however, this is, because of the movement peculiar to speed control, which takes time until the value is accumulated in the integrator of the speed controller.

Next, Fig. 7 and 8 show the results of experiments in which overcompensation and undercompensation were investigated by adjusting the factor  $\alpha$  of the magnitude of the dead-time compensation amount  $\Delta v$ . Fig. 7 shows the results of overcompensation when the compensation amount  $\Delta v$  is adjusted by  $\alpha = 1.25$ . Each waveform oscillates significantly at a frequency that depends on the speed, making the entire operation unstable.

Fig. 8 shows the result for the compensation amount  $\Delta v$  adjusted by  $\alpha = 0.75$ , which is the results of undercompensation. The system is able to start up without becoming oscillatory.

In the 5 to 6[s] of Fig. 5 to 8, the estimated primary current and secondary flux waveforms by the adaptive secondary flux observer change because the magnitude of the primary voltage error changes due to dead-time compensation. During 5 to 6[s], since the system is at a complete stop, the error information is not fed back to the observer and the same value is retained.

## 5. Dynamic Analysis by Linearized Model

To investigate under and overcompensation effects, dynamic analysis using a linearized model was performed by modeling the dead-time and its compensation.

### 5.1 Analysis Method

#### 5.1.1 Derivation of differential equations of the system

Simultaneous differential equations describing the same MRAS-based sensorless control system as in the experiment are derived using synchronous  $dq$  coordinate axes.

(I) Induction Motor, (II) Adaptive Flux Observer, and (III) Speed Estimator are as shown in equations (1), (3), and (4) in Chapter 2, respectively.

#### (IV) Automatic speed regulator (ASR)

ASR consists of a PI controller that considers the deviation between the speed command value  $\omega_{re}^*$  and estimated speed  $\hat{\omega}_{re}$  as the input. As the equation contains an integral, the order is increased by one.

$$i_{iq}^* = K_{PASR}(\omega_{re}^* - \hat{\omega}_{re}) + K_{IASR} \int (\omega_{re}^* - \hat{\omega}_{re}) dt \quad (8)$$

$$X_{\omega} = \int (\omega_{re}^* - \hat{\omega}_{re}) dt \quad (9)$$

#### (V) Automatic current regulator (ACR)

The current controller consists of a PI compensator for the deviation between the current command value and actual current. It also consists of an FF compensator for non-interference compensation.

This also has two more orders to express the integrator, as shown in equations (12) and (13).

$i_{iq}^*$  is the current command value with a first-order delay of the current control time constant, which is expressed in equation (14).

$$v_{1d}^* = R_{1s} i_{1d}^* - (\hat{\omega}_{re} + \hat{\omega}_{se}^*) \sigma_s L_{1s} i_{1qF}^* + K_{PACR}(i_{1d}^* - i_{1d}) + K_{IACR} \int (i_{1d}^* - i_{1d}) dt \quad (10)$$

$$v_{1q}^* = R_{1s} i_{1qF}^* + (\hat{\omega}_{re} + \hat{\omega}_{se}^*) L_{1s} i_{1d}^* + K_{PACR}(i_{1q}^* - i_{1q}) + K_{IACR} \int (i_{1q}^* - i_{1q}) dt \quad (11)$$

$$X_{id} = \int (i_{1d}^* - i_{1d}) dt \quad (12)$$

$$X_{iq} = \int (i_{1q}^* - i_{1q}) dt \quad (13)$$

$$\frac{d}{dt} i_{1qF}^* = \frac{1}{T_d} (i_{1q}^* - i_{1qF}^*) \quad (14)$$

#### (VI) Dead-time Compensation

The voltage error due to dead-time is compensated for by using the current command value. The compensating voltage is added to the voltage command value of the inverter. In reality, the polarity of the current command value is detected on the  $uvw$  3-phase stationary coordinates and compensated for the voltage command value of the same coordinates. In this analysis, the dead-time compensation voltage is expressed in the  $dq$  rotational coordinates using  $\Delta v_{dTc}, \Delta v_{qDTC}$  as a symbol. The voltage error caused by the dead-time in the inverter is compensated for by the compensation method described in Chapter 3. A compensation voltage in square wave of a magnitude of  $V_{DC} f_c T_{DT} \alpha$  [V] and having the same phase as that of the primary current command on the  $uvw$  3-phase stationary coordinates is added to the output voltage of the ACR<sup>(13)</sup>. When considered on three phases, the phase of the dead-time compensation voltage and the current command value are in the same phase. Therefore, the equation for the ratio in the first line of equation (15) is valid. The sum of the squares  $2\sqrt{2/3} V_{DC} f_c T_{DT} \alpha$  is converted to the same ratio as the current command of  $dq$  rotational coordinate. By solving this equation, equations (16) and (17) can be obtained. As the actual voltage is reduced by the dead-time, compensation voltage is added positively to the output voltage of the ACR, the voltage after compensation  $v_{1d}^*, v_{1q}^*$  is expressed as equations (18) and (19).

$$\left. \begin{aligned} \Delta v_{dDTC} : \Delta v_{qDTC} &= i_{1d}^* : i_{1q}^* \\ \sqrt{\Delta v_{dDTC}^2 + \Delta v_{qDTC}^2} &= 2\sqrt{\frac{2}{3}}(V_{dc} f_c T_{DT} \alpha) \end{aligned} \right\} \dots\dots\dots (15)$$

$$\Delta v_{dDTC} = \frac{2\sqrt{\frac{2}{3}}(V_{dc} f_c T_{DT} \alpha)}{\sqrt{1 + \left(\frac{i_{1q}^*}{i_{1d}^*}\right)^2}} \dots\dots\dots (16)$$

$$\Delta v_{qDTC} = \frac{2\sqrt{\frac{2}{3}}(V_{dc} f_c T_{DT} \alpha)}{\sqrt{1 + \left(\frac{i_{1d}^*}{i_{1q}^*}\right)^2}} \dots\dots\dots (17)$$

$$v_{1d}^* = v_{1d}^* + \Delta v_{dDTC} \dots\dots\dots (18)$$

$$v_{1q}^* = v_{1q}^* + \Delta v_{qDTC} \dots\dots\dots (19)$$

#### (VII) PWM Inverter

Voltage errors occur in inverters owing to the dead-time. A voltage error of an average magnitude of  $V_{DC} f_c T_{DT}$  [V] and having the same phase as that of the primary current on the  $uvw$  3-phase stationary coordinates is generated <sup>(13)</sup>. In this analysis, the voltage error is expressed in the rotational coordinates, as in the previous item of dead-time compensation, and is considered as equation (20) and (21). As the actual voltage is reduced by the dead-time, it is added negatively to the voltage command value after compensation, shown in equations (22) and (23).

$$\Delta v_{dDT} = \frac{2\sqrt{\frac{2}{3}}(V_{dc} f_c T_{DT})}{\sqrt{1 + \left(\frac{i_{1q}}{i_{1d}}\right)^2}} \dots\dots\dots (20)$$

$$\Delta v_{qDT} = \frac{2\sqrt{\frac{2}{3}}(V_{dc} f_c T_{DT})}{\sqrt{1 + \left(\frac{i_{1d}}{i_{1q}}\right)^2}} \dots\dots\dots (21)$$

$$v_{1d} = v_{1d}^* - \Delta v_{dDT} \dots\dots\dots (22)$$

$$v_{1q} = v_{1q}^* - \Delta v_{qDT} \dots\dots\dots (23)$$

#### (VIII) Mechanical System

The equation of motion of the mechanical system is expressed as follows: The load is assumed to be a constant torque load.

$$\frac{d}{dt} \omega_{re} = \frac{P_n}{J} \left\{ P_n \frac{M_m}{L_2} (\phi_{2d} i_{1q} - \phi_{2q} i_{1d}) - T_L \right\} \dots\dots\dots (24)$$

##### 5.1.2 Linearized Model

By considering infinitesimal variation of the nonlinear simultaneous differential equations derived in the previous section, a linearized model near the equilibrium point is derived. The model can be expressed as follows: The system matrix **A** and control matrix **B** are shown in the Appendix.

$$p\Delta x = A\Delta x + B\Delta \omega_{re}^* \dots\dots\dots (25)$$

$$\Delta x = \begin{bmatrix} \Delta i_{1d} & \Delta i_{1q} & \Delta \phi_{2d} & \Delta \phi_{2q} & \Delta X_{1d} & \Delta X_{1q} & \Delta \hat{i}_{1d} & \Delta \hat{i}_{1q} \\ \Delta \hat{\phi}_{2d} & \Delta \hat{\phi}_{2q} & \Delta X_{\omega} & \Delta i_{qF}^* & \Delta \hat{\omega}_{re} & \Delta \omega_{re} & \Delta X_{\omega} \end{bmatrix} \dots\dots (26)$$

#### 5.2 Analysis Results

The analysis was performed at low speed of  $\omega_{re}^* = 30$  [rad/s], approximately 1/10 of the rated speed, and the parameters used are listed in Table 2.

The equilibrium point in the analysis was the value obtained by solving the steady-state solution of the nonlinear simultaneous differential equations in the previous section using Newton's method in MATLAB. Table 3 shows each equilibrium point as an example.

Fig. 9 shows all the 15 poles at a compensation factor  $\alpha = 1.0$ . Complex conjugate roots are marked with the symbol “ ’ ”. Fig. 10 shows the trajectories of the poles when the dead-time compensation factor  $\alpha$  in equations (16) and (17) is varied from 0.0 to 1.25 in increments of 0.01. The starting point of the arrow was undercompensation and the ending point was overcompensation. The points of under compensation ( $\alpha = 0.75$ ), not excessive compensation ( $\alpha = 1.0$ ), over compensation ( $\alpha = 1.25$ ) are plotted in blue, red, yellow respectively. Of the poles shown in Fig. 10, only the pole closest to the origin was enlarged, as shown in Fig. 11. A

Table 3. Operating point for linearization

| Symbol               | Compensation amount |               |               |
|----------------------|---------------------|---------------|---------------|
|                      | 1.0                 | 0.75 times    | 1.25 times    |
| $i_{1d0}$            | 4.8100 A            | 4.8100 A      | 4.8100 A      |
| $i_{1q0}$            | 10.1995 A           | 11.2700 A     | 9.6321 A      |
| $\phi_{2d0}$         | 0.5964 Wb           | 0.5291 Wb     | 0.6450 Wb     |
| $\phi_{2q0}$         | 0.0000 Wb           | -0.0250 Wb    | 0.0268 Wb     |
| $\hat{i}_{1d0}$      | 4.8100 A            | 5.7062 A      | 3.8019 A      |
| $\hat{i}_{1q0}$      | 10.1995 A           | 11.2094 A     | 9.5470 A      |
| $\hat{\phi}_{2d0}$   | 0.5964 Wb           | 0.6109 Wb     | 0.5673 Wb     |
| $\hat{\phi}_{2q0}$   | 0.0000 Wb           | -0.0413 Wb    | 0.0479 Wb     |
| $\hat{\omega}_{re0}$ | 30.0000 rad/s       | 30.0000 rad/s | 30.0000 rad/s |
| $\omega_{re0}$       | 30.0000 rad/s       | 28.6357 rad/s | 30.7589 rad/s |
| $i_{1q}^*$           | 10.1995 A           | 11.2700 A     | 9.6321 A      |
| $i_{1qF0}^*$         | 10.1995 A           | 11.2700 A     | 9.6321 A      |

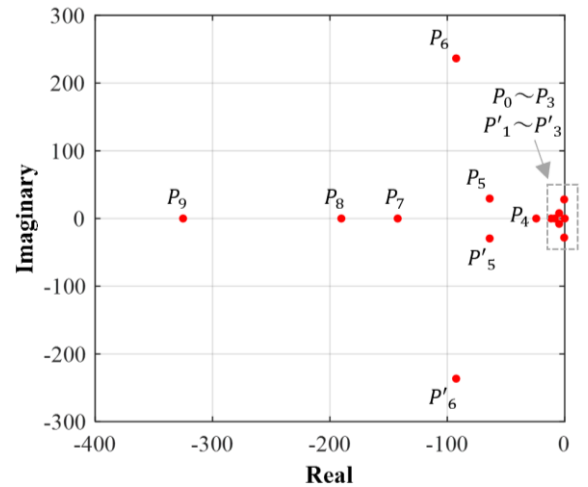


Fig. 9. All poles at normal compensation ( $\alpha = 1.0$ ) at

$$\omega_{re}^* = 30 \text{ [rad/s]}$$

typical example of undercompensation is when there is no dead-time compensation ( $\alpha = 0.0$ ). Fig. 11 shows that without dead-time compensation, the real part of the complex conjugate root of  $P_2$  becomes positive and it means unstable. However, in the experiment, the system is able to start up even without dead-time compensation without becoming oscillatory. This is because the speed command value is increased during startup, and the equilibrium point changes and becomes stable, enabling startup. For example, the pole at  $\omega_{re}^* = 50$  [rad/s] shown in Figure 12,  $P_2$  is on the stable side.

Since the analysis in this paper assumes that dead-time compensation is included originally, the poles only from undercompensation ( $\alpha = 0.75$ ) to overcompensation ( $\alpha = 1.25$ ) are shown in Fig. 13. The figure shows that as the compensation becomes overcompensated,  $P_1$  and its conjugate roots move to the unstable side. The angular frequency indicated by the imaginary part of this pole is almost identical to the angular frequency of the oscillation of each current, voltage, secondary flux, and also to the angular frequency that appears when speed command  $\omega_{re}^*$  is 30

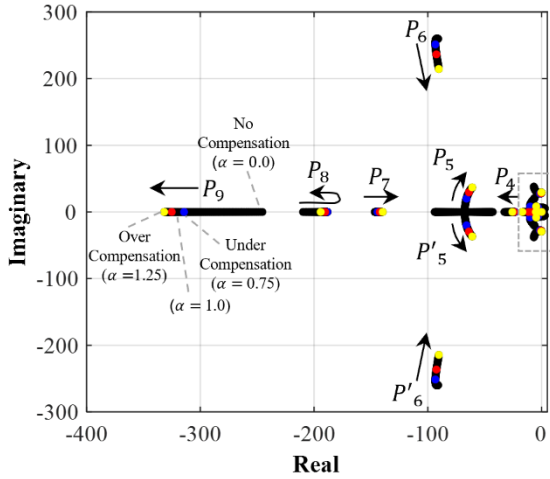


Fig. 10. All poles locus when  $\alpha$  is varied from 0.0 to 1.25 at  $\omega_{re}^* = 30$  [rad/s]

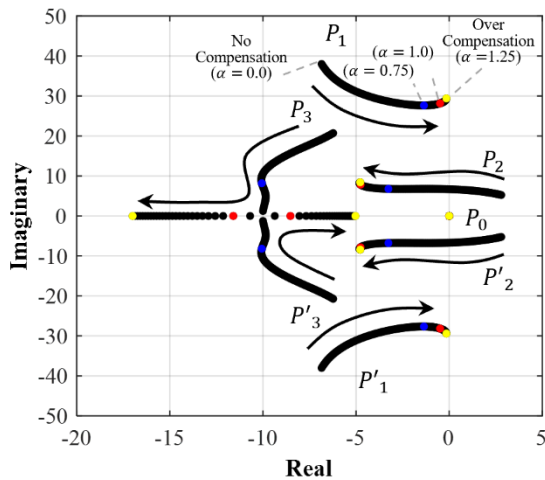


Fig. 11. Pole locus near the origin when  $\alpha$  is varied from 0.0 to 1.25 at  $\omega_{re}^* = 30$  [rad/s]

[rad/s] in the experimental results shown in Fig. 7, which indicate that the system is oscillatory owing to this pole. The experimental results in Fig. 7 show overcompensation, in which the estimated secondary flux oscillates, and as a result, the estimated speed

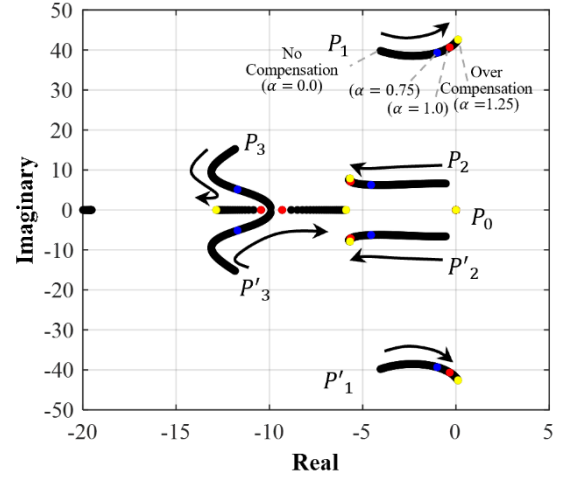


Fig. 12. Pole locus near the origin when  $\alpha$  is varied from 0.0 to 1.25 at  $\omega_{re}^* = 50$  [rad/s]

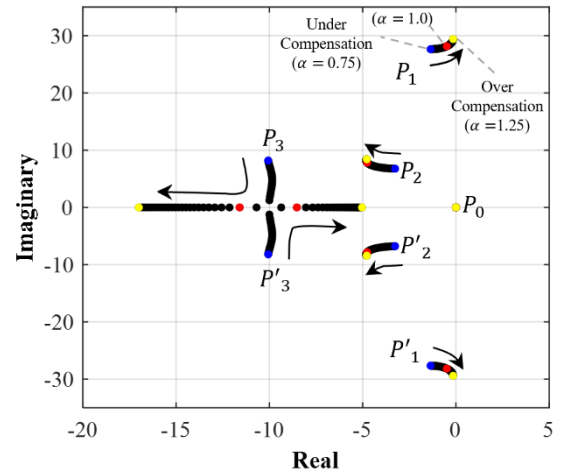


Fig. 13. Pole locus near the origin when  $\alpha$  is varied from 0.75 to 1.25 at  $\omega_{re}^* = 30$  [rad/s]

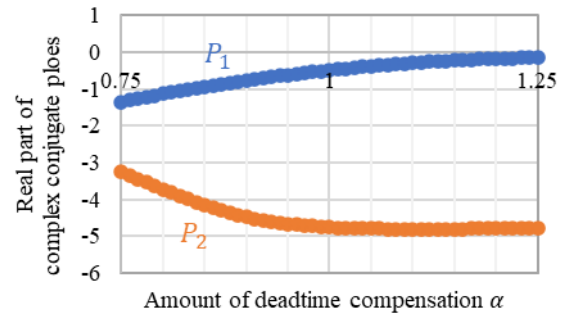


Fig. 14. Margin of stability due to the magnitude of the real part of the poles

calculated using this estimated secondary flux also oscillates. Consequently, the actual slip oscillates, causing oscillations in the real secondary flux. In addition, the speed electromotive force compensation term of the current control system, calculated using the estimated speed, oscillates, causing the actual primary current and secondary flux to oscillate, and the entire system becomes oscillatory.

The stability for different factor  $\alpha$  of the dead-time compensation was evaluated using the magnitude of the real part of the poles. Fig. 14 shows the real parts of the complex conjugate poles of  $P_1$  and  $P_2$ , the poles that are particularly close to the imaginary axis. As overcompensation is applied, the real part of  $P_1$  approaches the value of zero and the margin of stability becomes smaller.

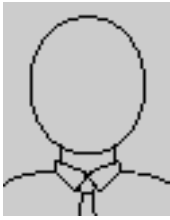
## 6. Conclusion

In this study, the MRAS-based secondary flux estimation method was applied as a measure of speed sensorless vector control method for an induction motor driver system. In this method, the primary voltage errors due to inverter dead-time, forward voltage drop of switching devices, and nonlinear characteristics affect the estimation of the secondary flux and primary current in the adaptive flux observer, degrading the speed estimation performance. Usually, FF dead-time compensation is performed; however, in practice, conventional dead-time compensation may not be proper because of the voltage error of the forward voltage drop and nonlinear characteristics of switching devices. First, the startup of the system is investigated experimentally when the compensation is changed from under-compensation to over-compensation and the behavior is then investigated by the pole configuration in the analysis. The results indicate that overcompensation is more prone to cause oscillations than undercompensation. In overcompensation, the secondary flux estimated oscillates depending on the angular speed, and the angular speed estimate obtained using this estimated secondary flux oscillates, causing the actual slip to oscillate. The non-interfering compensation term of the current control also becomes oscillatory and the real current and real secondary flux oscillate, and the entire system is considered oscillatory. Therefore, when using simple FF dead-time compensation, the amount of compensation should be set such that it is smaller than the actual amount, even if there are parameter variations.

## References

- (1) E. Izumi, M. Nagataki, K. Kondo, S. Tobayashi and H. Suzuki, "Improvement of Start-Up Performance at the Standstill Condition of Induction Motor Speed Sensorless Vector Control Using Adaptive Flux Observer" 2022 International Power Electronics Conference (IPEC-Himeji 2022), (2022)
- (2) K. Kondo and K. Yuki, "An application of the induction motor speed sensor less control to railway vehicle traction system," Conference Record of the 2002 IEEE Industry Applications Conference. 37th IAS Annual Meeting (Cat. No.02CH37344), vol.3, pp. 2022-2027 (2002)
- (3) T. Ogawa, S. Ishida, T. Kojima, T. Sato, H. Taguchi and S. Ohashi, "Speed sensorless vector control for rolling stock," 2005 European Conference on Power Electronics and Applications, pp. 7 (2005)
- (4) Geng Yang and Tung-Hai Chin, "Adaptive-Speed Identification Scheme for a Vector-Controlled Speed Sensorless Inverter-Induction Motor Drive", IEEE Trans., vol.29, no.4, pp.820-825 (1993)
- (5) H. Kubota, "DSP-Based Speed Adaptive Flux Observer of Induction Motor" IEEE Trans., vol.29, no.2, pp.344-348, 1993
- (6) N. Kawamura, T. Zanma, K. Koiwa, Kang-Zhi Liu and M. Hasegawa, "Simultaneous Estimation of Rotor Speed and Stator Resistance in Speed-Sensorless Vector Control of IMs," IEEE Journal of Industry Application, Vol.10, No.4 pp469-480 (2021)
- (7) T. Hoshino, and J. Itoh, "Output voltage correction for a voltage source type inverter of an induction motor drive," IEEE Trans. Power Electronics, vol. 25, pp. 2440-2449 (2010)
- (8) Y. Park, and S. Sul, "A novel method utilizing trapezoidal voltage to compensate for inverter nonlinearity," IEEE Trans. Power Electronics, vol. 27, pp. 4837-4846 (2012)
- (9) R. Raute, and C. Caruana, "Analysis and compensation of inverter nonlinearity effect on a sensorless PMSM drive at very low and zero speed operation," IEEE Trans. Industrial Electronics, vol. 57, pp. 4065- 4074 (2010)
- (10) S. Wei, Y. Yong, W. Gaolin and X. Dianguo, "Accurate inverter error compensation using self-tuning stator current estimation error in sensorless induction motor drives," 2014 IEEE Industry Application Society Annual Meeting, pp. 1-6 (2014)
- (11) Li Ding and H. Sugimoto, "A Consideration and Improving Method about Stability of Vector Controlled Induction Motor System without Speed Sensor Driven with Controlled Voltage Source Using Adaptive Secondary Flux Observer", IEEE Trans., vol.120, no.10 pp.1225-1234 (2000) (in Japanese)
- (12) H. Kubota and K. Matsuse, "Speed Sensorless Field-Oriented Control of Induction Motor with Rotor Resistance Adaptation", IEEE Trans., vol.30, no.5 pp.1219-1224 (1994)
- (13) T. Hoshino, J. Itoh and T. Kaneko, "Dead-Time Voltage Error Correction with Parallel Disturbance Observers for High Performance V/f Control," IEEE Industry Applications Annual Meeting, pp. 2038-2044 (2007)



**Erina Izumi**

(Student Member) received the B. E. degree from School of Advanced Science and Engineering, Department of Electrical Engineering and Bioscience, Waseda University, Japan, in 2021. She is currently a student at the graduate school of the same university. Her main research interests include speed sensorless control of induction motor.

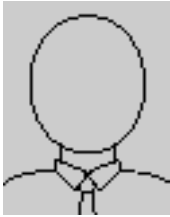
**Masaki Nagataki**

(Member) received the M. E. degree from Graduate School of Science and Engineering, Chiba University, Japan, in 2020. He is currently pursuing a Ph.D. at the Graduate School of Advanced Science Engineering, Waseda University. His research interests include power electronics and motor-drive.

**Keiichiro Kondo**

(Senior Member) received B.S. and Dr. Eng. from Faculty of Electrical Engineering, Department of Science and Technology, Waseda University, in 1991 and 2000, respectively. He was with Railway Technical Research Institute from 1991 to 2006 and was engaged at R&D for power electronics applied railway vehicle traction system. From 2007 to 2018, he was in Electrical and Electronic Engineering Course of Graduate School of Chiba University. Since 2018, he is a professor at School of Advanced Science and Engineering, Department of Electrical Engineering and Bioscience, Waseda University. His current research interests are power electronics, AC motor drive, energy storage devices, wireless power transmission and their applications to railway systems. Prof. Kondo is titled Professional Engineer in Japan (Mechanical Engineering Technical Management). He is a Member of IEEE.

**Shunsuke Tobayashi** (Member) received the B.S. and M.S. degrees in electrical and electronic engineering from Chiba University, Japan, in 2008 and 2010 respectively. Since 2010, he has been working in Toshiba Mitsubishi-Electric Industrial Systems Corporation, Tokyo, Japan. From 2012 to 2013, he was a visiting researcher at University of Wisconsin-Madison, WI, USA. He currently develops power electronics systems for motor drive, particularly medium-voltage class for industrial applications.

**Hiromitsu Suzuki**

(Member) received the B.S. and M.S. degrees in information technology from Okayama University, Japan, in 1996 and 1998 respectively. He was in Toshiba Corporation from 1998 to 2005 and had been engaged in development of drive systems. Since 2005, he has been with Toshiba Mitsubishi-Electric Industrial Systems Corporation, Tokyo, Japan. He is currently engaged in drive system development for general industrial applications.